

Project Plan

Full Unit – Project Plan

Student Support and Development Report | Building Mobile
Application

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A report submitted in part fulfilment of the degree of

BSc (Hons) in Computer Science

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October 09, 2025

1 Abstract

Starting university or leaving secondary education is typically of a moment of high stress and development for young people as they are learning to navigate the world without the structure of most secondary education systems. There is a significant decrease in direction given to students and for most this is first time that they will living alone and away from family. This can lead to many students' developing poor mental health and struggle to find effective coping mechanisms that can support them^[2]. Student mental health services have seen an increase in students requiring support and an increase of students' report feeling depressed, anxious and experience high levels of stress^[3]. Many contribute these feelings towards stressors such as financial inexperience, educational demands and poor organisational skills. Many platforms require significant input from users, are difficult to set up or subscriptions are needed to be paid for them to be effective.

If someone were to already be overwhelmed, facing financial insecurity or unable to dedicate time to set up the platform, this ultimately negates the point of service. Instead, I am going to provide an app to users where they can stay organised and learn effective life skills while not being burdened.

To achieve this, I am going to develop an app that introduces easily accessible progress feedback areas, promoting good lifestyle habits, and understanding how certain actions can lead to burnout. Some healthy levels of stress can help promote better long-term memories and support the learning of students, so encouragement to try new skills and techniques can be added through memory logs and achievements^[1]. We can also monitor to track these same actions and see where harmful habits are created, for example, students studying for extreme periods of time without breaks. I hope to achieve this by using GPS features to allow for 'study spaces' to be determined by users, allowing them to automatically track study periods but also be alerted when to take a break. This will be done through a combination of the user's GPS tracker and a timer that will log a session if the user is within a saved 'study space' for longer than 5 minutes. Libraries and other common studying spaces can be included if users do not want to create their own 'study spaces'.

Users will also have access to smart planners that can have automatic and manual entries added to help them with possible deadlines. These can be synced with pre-existing calendars and used in conjunction with the journal log. To support reflection and consistency, user will have the option to include a daily log in an audio or written format. Photos taken throughout the day can be included as well, with an option to see where they were taken on a map. And at the end of each week, a summary of their achievements and progress will be given, further enforcing those positive habits.

Users will also have the option to track their budgeting via smart categories and alerts once thresholds are reached, with an emphasis on visual representations to help convey growth and possible improvements.

Market research and testing will also be carried out throughout to see how users utilise the software, and what features of the UI are easily understandable. This should be an app for beginners and should be easy to approach.

This app will be developed using the Android Studio IDE using the Kotlin programming language. Testing will be done through the built-in testing platform in Android Studio, using a TDD approach to ensure all features work as expected. There will be a round of further security and functional testing completed to ensure users are able to use the app safely. Figma will be used for UI design and visual planning.

Overall, while there exists platforms for students and people in their early careers, I hope to provide an option to help beginners make decisions to support their health and learn new things. Existing platforms can be stage after this app, as this app should help users find works best for them and let them easily keep track of their progress.

2 Timeline

Initially, I would like to focus on getting the core basic functionality to be completed during term 1 before moving onto any fine tuning of features or major changes made to the UI. This allows time for further research and testing to happen alongside the development.

2.1 Term 1

Week	Primary Deliverable	Includes	Achievement	PERT Value (H)
1-2 (22/09 – 05/10)	Initial Research	Understanding the project scope and requirements, Initial research	Brainstormed project ideas, finding reliable sources	M(8), O(10), P(15) - 10
3-4 (06/10 - 19/10)	Initial project plan, App's front page, App life cycle plan, Basic UI Navigation	Provide an appropriate plan for development, understanding how to use and test in Android Studio, Basic UI that can be used to build functionality, provide a better proof of concept	Breakdown of major features into subtasks, Criteria for success, Further research, workable project, workable front page, Plan of App flow, Functioning UI with minimal features	M(10), O(15), P(30) - 16

5-6 (20/10 – 02/11)	Implementing GPS tracking and creating 'study spaces'	Tracking the user via GPS, recording specific coordinates, recognising users within recorded 'study spaces', unit testing	Allow users to be tracked using GPS and recognise 'study spaces'	M(15), O(20), P(35) - 22
7 (03/11 - 09/11)	Implement user accounts	Create user accounts, associate saved 'study spaces' with users, securely store sensitive information, unit testing	Secure general user accounts	M(8), O(10), P(15) - 10
8 (10/11 - 16/11)	Implement basic planner feature	Calendar view, Task view, Event view, Sync with existing Calendars, Categories, Manual entries, unit testing	Create a simple to use drag and drop calendar	M(8), O(10), P(15) - 10
9 (17/11 - 23/11)	Testing and debugging	All code is fully tested and debugged, Security testing, system testing, functional testing	Stable version of all working code	M(8), O(10), P(15) - 10
10-11 (24/11 - 07/12)	Prepare for interim report	Interim Report, Presentation	Complete interim report	M(15), O(20), P(35) - 22

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3 Risks and Mitigations

3.1 Hardware Failure

There is always a risk of a hardware failure in the middle of development. At any stage, this could significantly affect progress and prevent deadlines from being reached. While the likelihood of this occurring is minimal, the impact would be significant if not catastrophic, especially further down the timeline. To prevent this, I will be keeping my code and research on GitLab and other cloud-based platforms. I will also be keeping a version stored to a physical disk drive. This also does not limit development to my local machine and can be developed elsewhere. While there is still a possible risk for reoccurrence or copies being lost, having multiple copies should allow for one to always be available.

3.2 Poor Estimation of Tasks

Due to the length of the project, there is minimal room for errors in the period spent on each task. If specific features were to take longer to the point where it overruns into other deadlines, this could create a knock-on effect that derails the whole plan. To mitigate this, splitting each task into its components and using a multi-factor estimation method, and then later using the historic data to evaluate the best course of action should reduce the likelihood of this occurring. Any features that have a dependency will also be addressed early on, allowing for concurrent development to exist if required.

3.3 Balance between the Report and Code

There is a significant risk that the app development can overtake the completion of the report. This could make documentation more difficult to revisit as past actions could be forgotten. To mitigate this, I will be maintaining my diary and workbook that will have consistent dated updates. I will also be dedicating a timetabled slot every week for specifically the report, so it does not get overshadowed completely by development. I will be honest in my reflections and put into perspective the progress made.

3.4 End-User Safety

Users can be tracked via their GPS and account details can be shared amongst peers. This could introduce malicious actors who could greatly harm users. While the likelihood is

minimal, the impact could be detrimental to users' safety and well-being. All sensitive features such as GPS and public information will be highlighted to users and can be turned off at any time. Data will be stored securely, with personal data likely being stored locally to the user, and users will have limited access to sharing information. This will not completely negate possible risk but makes it harder for malicious actors to possibly gain access.

3.5 Sensitive Data Collection

Many of the user would be sharing account and sensitive personal data with the platform, and if this were to be compromised could be extremely dangerous for users. Similarly to the end-user's safety, while this would be unlikely, the impact would be significant. To mitigate this, details of users will be encrypted and stored securely, passwords will not be stored in the clear and strict access controls will be implemented. Any templates will be stored separately from any user databases to avoid accidental cross over.

3.6 Scope Creep

Scope creep can lead to a difficulty in understanding when to stop adding features, which can result in missed deadlines or underdeveloped features. To prevent this, I will be making a clear plan of what I want to achieve and adhering to a strict schedule. After the first term, I will re-evaluate the progress made and use that evidence to plan the second term. Features will also be given a priority allowing for opportunities for non-essential features to be removed in favour of not compromising the overall quality.

Acronyms

PERT – Program [Evaluation and Review Technique](#) - A formula $((\text{Optimistic Estimate} + 4 * \text{Most Likely Estimate} + \text{Pessimistic Estimate}) / 6)$ to estimate the length of time required for the development of a feature.

TDD - Test-driven development

Glossary

Study spaces - Areas that students have predetermined as the places they consistently study, possibly only want to . They will have the option to automatically select libraries as existing options.

Bibliography

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 - Utilising the use of healthy levels of stress to promote positive habits and study periods, while reducing potential of burnout and excessive stressors.
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 - Understanding contributing factors to excessive stress.
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 - Researching into different visual cues given to student to improve task completion.
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 - Finding ways that mobile and social learning are used to help students retain information.
- [7] Oliveira, W., Hamari, J., Joaquim, S. et al. The effects of personalized gamification on students' flow experience, motivation, and enjoyment. *Smart Learn. Environ.* 9, 16 (2022). <https://doi.org/10.1186/s40561-022-00194-x>
 - Introducing elements of positive reinforcement to encourage habits to form.
 - Using methods that adjust to feel manageable and achievable to users, and making the habits stick for the long-term.
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 - Looking into how seeing how progression prompts further positive reinforcement.
 - Continuing to find ways that can help students feel accomplished and motivated.